

(d) Argument FOR:

- Increasing expenditure on infrastructure can increase the capital endowment in the future, raising the aggregate output in the long run. (2)

Argument AGAINST:

- Increasing expenditure on infrastructure would raise the US Government's expenses, worsening the fiscal balance. (2)

4.4 DEPOSIT CREATION

MARKING SCHEME

1990/CE/II/45 C	1996/CE/II/36 B	2001/CE/II/26 C	2005/CE/II/37 A (57%)	2010/CE/II/35 B (68%)
1991/CE/II/31 B	1997/CE/II/33 B	2002/CE/II/35 C (84%)	2005/CE/II/50 B (46%)	2010/CE/II/36 C (43%)
1991/CE/II/32 B	1997/CE/II/34 A	2002/CE/II/36 D (56%)	2006/CE/II/36 D (61%)	2012/DSE/I/27 C (71%)
1992/CE/II/34 B	1998/CE/II/33 A	2002/CE/II/37 C (59%)	2006/CE/II/37 B (57%)	2013/DSE/I/26 C (58%)
1992/CE/II/35 D	1998/CE/II/34 D	2002/CE/II/38 A (25%)	2006/CE/II/38 C (37%)	2014/DSE/I/30 D (56%)
1992/CE/II/32 C	1998/CE/II/52 C	2003/CE/II/34 D (17%)	2007/CE/II/36 D (49%)	2016/DSE/I/31 D (65%)
1994/CE/II/40 A	1999/CE/II/36 B	2003/CE/II/35 B (24%)	2008/CE/II/38 B (39%)	2016/DSE/I/33 A (deleted)
1994/CE/II/42 D	1999/CE/II/38 A	2003/CE/II/36 C (16%)	2008/CE/II/39 D (60%)	2017/DSE/I/31 B (36%)
1995/CE/II/42 B	2000/CE/II/29 C	2004/CE/II/38 A (73%)	2009/CE/II/38 A (44%)	2019/DSE/I/28 D
1996/CE/II/35 C	2000/CE/II/30 B	2004/CE/II/39 B (34%)	2009/CE/II/39 D (37%)	2020/DSE/I/33 B
2022/DSE/I/30 D	2022/DSE/I/32 C	2023/DSE/I/34 C	2024/DSE/II/34 B	2025/DSE/II/34 D

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1990/CE/2/(d)(ii)	
(I) $\$10\,000 \times (1 - 25\%)$	(2)
= \$7 500	(1)
Possible way: to recall loans / sell investment assets	(2)
(II) $\$10\,000 \times (1 / 25\%)$	(2)
= \$40 000	(1)
1991/CE/I/5(b)	
(i) $\$200 / \$800 = 25\%$	(2)
(ii) Actual reserves – (Deposits $\times$ Required reserve ratio)	
$\$50 - (\$100 \times 25\%) = \$25$	(2)
(iii) Excess reserves $\times$ Banking multiplier	
$\$25 \times (1 / 25\%) = \100	(3)
Assumptions:	
- no cash drain	(2)
- people are willing to borrow	(2)

1992/CE/I/3(b)(ii)

- 100% reserves required by the government
 - the bank keeps all the \$1000 as reserves
 - no one borrows from the bank
 - any cash that leaks out of the bank is not redeposited with the banking system
- (Mere listing of points without elaboration - max. 2 marks.)
[Mark the **FIRST TWO** points only.]

(3@, max: 6)

1993/CE/I/5(c)(ii)

$$\begin{aligned} &\$3 \text{ million} \times (1 / 25\%) \\ &= \$12 \text{ million} \end{aligned}$$

(2)
(1)

1994/CE/I/5

$$\begin{aligned} &\text{Maximum amount of deposits to be created} = \$100 \times (1 / 20\%) \\ &= \$500 \end{aligned}$$

(3)
(1)

1995/CE/I/10(a)(iii)

Brief description of the changing process in a correct sequence:

Initial withdrawal of \$M $\Rightarrow$

- (Step 1:) Banks lose reserves (i.e. actual reserves < required reserves)
- (Step 2:) Banks call back loans / sell bank assets to the public
- (Step 3:) Further withdrawal of bank deposits to repay the loans / to buy bank assets
The process of bank deposit contraction goes on and on
- (Step 4:) Ultimately, total bank deposits contract
by $\$M \times (1 / 20\%) = \$5M$

(2)
(2)
(2)
(1)
(1)
(2)
(max: 8)

1996/CE/I/11(c)

- (i) No change in total bank deposits, because
it is only a switch from one's bank deposit to another's.

(1)

OR

there may be some deposit contraction at first, but then it would be offset by the deposit expansion later on.

(2)

- (ii) Banks gain excess reserves

$$\begin{aligned} &\Rightarrow \text{increase in total bank deposits} \\ &\text{by an amount of } \$5\text{mn} \times (1 / 25\%) = \$20\text{mn} \end{aligned}$$

(1)
(1)
(2)

1997/CE/I/9(c)

(A logical sequence of the following is required.)

An injection of new money (e.g. the initial \$10 b.) to the local banks

(1)

$\Rightarrow$ banks gain excess reserves (\$R)

(1)

and will loan out the excess reserves

(1)

$\Rightarrow$ the loan will be redeposited into the banking system

(1)

But due to the following factor(s), the redeposit amount is smaller than SR:

- initially not all the new money (the \$10b.) are deposited to the local banks
- local banks hold excess reserves
- insufficient demand for loan
- cash drain

(2@, max: 4)

Elaboration on how any one of the above factors would lead to a total bank deposit amount smaller than the maximum amount

(2)
(max: 8)

1998/CE/I/9(a)

- (Step 1:) After the bank accepts Mr Wong's \$1 m deposit, it will lend out / invest 75% or \$0.75 m in the first round
- (Step 2:) The \$0.75 m loan / investment will be redeposited into the banking system
- The process of (Step 1 - Step 2 - Step 1...) will go on and on until
finally the total deposit created = initial deposit $\times$ (1 / reserve ratio)
 $= \$1 \times (1 / 25\%) = \4 m

(2)
(2)
(2)
(2)

1999/CE/I/6(a)

$$\begin{aligned} &\$10 \text{ 000} + [\$10 \text{ 000} \times (1 - 25\%)] \\ &= \$17 \text{ 500} \end{aligned}$$

(2)
(2)

2000/CE/I/10(a)

Answer 1

(Assume that the public does not use the \$50 mn cash in hand to repay bank loans when the banks call back loans due to the withdrawals)

$$\begin{aligned} \text{(i)} \quad &-\$10 \text{ mn} \times [1 - (200 / 1 \text{ 000})] \\ &= -\$50 \text{ mn} \end{aligned}$$

(2)
(1)

$$\begin{aligned} \text{(ii)} \quad &\text{Change in money supply} \\ &= \text{Change in cash in hand} + \text{Change in bank deposits} \\ &= (+\$10 \text{ mn}) + (-\$50 \text{ mn}) \\ &= -\$40 \text{ mn} / \text{a contraction of } \$40 \text{ mn} \end{aligned}$$

(1)
(2)
(1)

Answer 2

(Assume that the public considers to make use of the \$50 mn cash in hand to repay bank loans when the banks call back loans due to the withdrawals)

- (i) The max. change is the same as the one in Answer 1. However, the actual change depends on how much of the \$50 mn cash in hand is used to repay the bank loans. The min. change is -\$10 mn. (Max: 3)
- (ii) Use the same formula in Answer 1. The change in bank deposits varies from a max. decrease of \$50 mn to a min. decrease of \$10 mn, depending on how much of the \$50 mn cash in hand is used to repay the bank loans. The change in cash in hand varies from a max. to +\$10 mn to a min. of +\$2 mn. Therefore, the final numerical answer varies from -\$40 mn to -\$8 mn. (Max: 4)

2001/CE/I/10(b)

$$\begin{aligned} \text{(i)} \quad &\text{Actual reserve ratio} = 250 / 1 \text{ 000} \\ &= 0.25 \\ &\text{Excess reserve} = (0.25 - 0.2) \times \$1 \text{ 000 mn} \\ &= \$50 \text{ mn} \end{aligned}$$

(1)
(1)
(1)
(1)

- (ii) The withdrawal reduces the amount of reserves to \$210 mn.
The new amount of reserves can back up \$210 mn $\times$ (1 / 20%)
 $= \$1 \text{ 050 mn}$ of deposits

(1)
(2)
(1)

2002/CE/I/10(a)

$$\begin{aligned} \text{(i)} \quad &\text{The required reserve ratio} = (20 / 100) = 0.2 \text{ or multiplier} = 5 \\ &\text{Max. amount of deposits created} = \$10 \text{ m} \times 5 = \$50\text{m} \\ &\text{Max. amount of money supply created} = \text{Max. amount of deposits created} - \text{decrease in cash in public circulation} \\ &= (\$50\text{m} - \$10\text{m}) = \$40 \text{ m} \end{aligned}$$

(1)
(1)
(1)
(2)

- (ii) - there is cash leakage
- the banks decide to hold excess reserves
- insufficient demand for loans
- [Mark the **FIRST TWO** points only.]

(2@, max: 4)

2003/CE/I/11

- (b) (ii) Some banks cannot lend out their excess reserve as the demand for bank loans ↓
 Step 1: loans that can be made by banks would decrease (1)
 Step 2: money re-deposited into the banking system would decrease (1)
 Correct sequence : Step 1 → Step 2 (1)
 Therefore, deposit creation ability ↓ (1)

- (c) Legal reserve ratio = 0.2 (1)
 Max. deposit = \$250 m × (1 / 20%) (1)
 = \$1 250 m (2)

2004/CE/I/6(b)

- Initial withdrawal of deposits, (1)
 ⇒ banks lose reserves / insufficient reserves / actual reserves smaller than required reserves (1)
 ⇒ banks have to call back loans / sell assets to the public (1)
 ⇒ further withdrawals of deposits (1)
 The process goes on and on. (1)

2005/CE/I/5

- (a) The amount of excess reserve (1)
 = \$[120 - (1 000 × 10%)] m = \$20 m (1 + 1)
- (b) Maximum possible amount of bank deposits (1)
 = \$120 m × (1 / 10%) = \$1 200 m (1 + 1)

2006/CE/I/6

- (a) 10% (1)
- (b) Yes because (1)
 when total deposits reduce to \$1 150 (= \$1 500 - \$350),
 it only needs a reserve of \$115 (= \$1 150 × 10%) to back up the deposits.
 Since the reserve assets only reduce to \$150 (= \$500 - \$350), there is still an excess reserve of \$35. (2)

2007/CE/I/7

- (a) \$200 (1)
- (b) (i) -\$100 (1)
- (ii) -\$30 (2)

2008/CE/I/10

- (a) An increase in the reserve ratio (1)
 ⇒ insufficient reserves / actual reserves smaller than required reserves (1)
 ⇒ banks have to call back loans / sell assets to the public (1)
 ⇒ withdrawals of deposits (1)
 The process will go on and on (until the actual reserve is equal to the required reserve).* (1)
 [*Remark: This point deserves one mark only when the sequence of the above steps is correct]

- (b) New required reserve ratio = 0.2 + 0.05 = 0.25

Amount of bank deposits that \$1 000 million reserve can back up = \$1 000 million × (1 / 25%) (1)
 = \$4 000 million (1)

OR
 Change of bank deposits = -\$250 million × (1 / 25%) (1)
 = -\$1 000 million (1)

Change in MS = Change in Cp + change in Dd / No change in Cp / MS = Cp + Dd (1)
 Change in MS = -\$1 000 million (1)

(MS = money supply;
 Cp = cash in the hands of the non-bank public; and
 Dd = bank deposits.)

2009/CE/I/6

- (a) The minimum required reserve is \$1 000 × 20% = \$200 (1)
 Excess reserve = \$100 (1)
- (b) Total amount of deposits at maximum = \$300 × (1 / 20%) (1)
 = \$1 500 (1)
OR
 Amount of extra deposits created = \$100 × (1 / 20%) (1)
 Therefore, total amount of deposits at maximum = \$1 000 + \$500 = \$1 500 (1)

2010/CE/I/11(b)

- (i) Banking multiplier = 1 / 40% = 2.5 (1)
 The max. amount of deposits that can be supported by \$2 000 million reserve at 40% reserve ratio
 = \$2 000 mn. × (1 / 0.4)
 = \$5 000 mn. (1)
 Therefore, the max. change in bank deposits = \$5 000 mn. - \$8 000 mn. = -\$3 000 mn. (1)
- (ii) An increase in the reserve ratio (1)
 ⇒ the actual reserve is smaller than the desired reserve (1)
 ⇒ banks have to call back loans / sell assets to the public (1)
 ⇒ withdrawals of deposits (1)
 The process will go on and on (until the actual reserve is equal to the desired reserve).* (1)
 [*This 4th mark will be awarded only if the above three marks are already awarded.]
 [#If the process starts with "withdrawals of deposits", then award only a maximum of 2 marks]

1997/AL/II/9

- (a) reserve ratio = 0.25

- (b)

Assets (\$)		Liabilities (\$)	
Reserves	1 000	Deposits	5 500
Loans	4 500		

When the required reserve ratio is 10%, the maximum loan the banking system can extend is \$9 000. Since the maximum demand for loan is \$4 500, the loan extended will be \$4 500. Money supply is \$5 500.

- (c) A government can restrict the maximum amount a banking system can lend, or the maximum money supply. But the actual amount of lending or the actual money supply is also affected by the demand for loan.

2022/DSE/IB/8

(a) 25%.

(b) Original money supply = $4000 + 1000 = 5000$ (\$million)
 New deposit = $(1000 + 600) \times \frac{1}{25\%} = 6400$ (\$ million)
 New money supply = $6400 + 400 = 6800$ (\$million)
 Change in Ms = $6800 - 5000 = 1800$ (\$million)

2023/DSE/II/10i, ii

(b) (i) Expected real rate of return = $4.5\% - 4.5\% = 0\%$ (1)
 Actual real rate of return = $4\% - (-1\%) = 5\%$ (1)
 (ii) Gain, because the actual real rate of return is higher than the expected real rate of return. (1)

2024/DSE/II/8

(a) The required reserve ratio = $[(\$600 \text{ million} + \$100 \text{ million}) / \$3 \text{ 500 million}] \times 100\% = 20\%$ (1)
 (b) The new required reserve ratio = $20\% - 5\% = 15\%$ (1)
 The maximum amount of deposits after deposit creation = $\$600 \text{ million} / 15\% = \4 000 million (1)
 The maximum change in deposits = $\$4 \text{ 000 million} - \$3 \text{ 500 million} = \500 million (1)

1998/AL/II/9

(a) Money supply = Deposits + amount of currency held by the public
 = $\$4 \text{ 000} + \$500 = \$4 \text{ 500}$

(b) (i)

Assets (\$)		Liabilities (\$)	
Reserves	1 100	Deposits	4 100
Loans	3 000		

Money supply = $\$4 \text{ 100} + \$400 = \$4 \text{ 500}$

(ii)

Assets (\$)		Liabilities (\$)	
Reserves	1 100	Deposits	4 400
Loans	3 300		

Money supply = $\$4 \text{ 400} + \$400 = \$4 \text{ 800}$

(c) It will increase the money supply only if the banks can lend out some or all of their excess reserves.

2017/DSE/I/8

(a) $RRR = (\$1 \text{ 000} - \$250) / \$3 \text{ 000} = 0.25$ (1)

(b) The change in deposits = $\$250 \times (1 / 0.25) = \1 000 (2)

2019/DSE/II/7

(a) Excess reserves = $\$300 \text{ million} - \$1000 \text{ million} \times 20\% = \100 million (1)

(b) Old money supply = $\$1000 \text{ million} + \$200 \text{ million} = \$1200 \text{ million}$ (4)
 New money supply = $(\$300 \text{ million} + \$200 \text{ million}) \times \frac{1}{20\%} + \$0 \text{ million} = \$2500 \text{ million}$
 Maximum possible change in the money supply = $\$2500 \text{ million} - \$1200 \text{ million} = \$1300 \text{ million}$

2021/DSE/II/6

6. (a) Required reserve ratio = $(600 - 300) / 1500 = 20\%$ (1)

(b) New required reserve ratio = $20\% - 5\% = 15\%$ (3)
 New deposit = $1500 + (600 - 0.15 \times 1500) / 0.15 = 4000$ (\$million)
 New money supply = $4000 + 100 = 4100$ (\$million)[†]

In real-world practice of monetary policy, adjustment in the required reserve ratio (RRR) is understood as a change in percentage point or basis point (i.e. a level change in RRR) rather than a percentage change in RRR. The following answers are also accepted in case the candidate (mis)understood the 5% reduction in RRR as a percentage change rather than change in percentage point.[‡]

New required reserve ratio = $20\% \times (1 - 5\%) = 19\%$
 New deposit = $\$3157.89 \text{ million}$
 New money supply = $\$3257.89 \text{ million}$